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A Comparative Analysis of 4-Bit Adder Architectures for Enhanced Efficiency in Conventional CMOS Technology

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***Abstract*—Throughout the years, with the miniaturization of VLSI technology, adders form an essential component of almost every contemporary integrated circuit. The prerequisite of an adder is that it is primarily fast and secondarily efficient in terms of power consumption and chip area. The race to create the most optimized device structure is getting harder and faster, as more compact digital devices with the lowest power consumption keep coming out and replacing each other. The optimal adder architecture is crucial for modern digital systems to enhance performance. Hence, choosing a perfect architecture that is compact and has low power consumption can be done with conventional CMOS technology by rationally comparing the different types of adders based on various performance parameters. Among the vast realm of adders, this research paper analyzes five 4-bit adder architectures, namely Ripple Carry Adder (RCA), Carry Look Ahead Adder (CLA), Carry Select Adder (CSA), Carry Save Adder (CSaA), and lastly Carry Skip Adder (CSkA). The study evaluates performance metrics of adder architectures including transistor count, average power, area, frequency, and propagation delay using Cadence Virtuoso. This work shows that RCA has simplicity but a significant propagation delay. On the other hand, the CLA improves speed but uses increased area. Innovative carry-save techniques can help to mitigate delay. Considering all parameters, CSkA is the most efficient adder with the least transistor count of 182, area of 16880 μm^2, and average power requirement of 50.04 μWatt.**

***Index Terms*—Adder, Carry Look Ahead Adder (CLA), Carry Select Adder (CSA), Carry Save Adder (CSaA), Carry Skip Adder (CSkA), Delay, Frequency, Power, Ripple Carry Adder (RCA).**

I. Introduction

Modern electronic devices require higher performance, speed, and power with lower input power, requiring precision in ICs and portability for multiple applications which simultaneously increases the challenges for modern-day circuit designers [1]. In modern microprocessors, adders play a crucial role in completing basic arithmetic logical tasks such as addition, subtraction, and multiplication [2], [3]. As an essential component of digital circuits, adders have seen modifications to their designs throughout time in response to the escalating performance and power efficiency prerequisites. It is well known that the area and delay are inversely connected, thus studies are being conducted to adapt newer logic modifications for adders with a lower area [4]. In brief, the growing demand for faster and power efficient digital circuitry has propelled the growth of adders. As digital circuits become ever more complicated, newer adder designs will eventually be mandated to match their power and performance specifications [5]. A half adder is a basic building component in digital circuits that adds two single-bit binary values. It does simple addition and generates SUM and CARRY. It requires one XOR gate and one AND gate (Eqtn. 1, 2). A full adder is built using two half adders in cascade form along with an extra OR gate. It is an extension of half adders which is a powerful tool in digital circuits for adding three bits of numbers. It also produces SUM and CARRY (Eqtn. 3, 4) [6].

$$SUM = A \oplus B \tag{1}$$

$$CARRY = A \bullet B \tag{2}$$

$$SUM = A \oplus B \oplus C \tag{3}$$

$$CARRY = A \bullet B + B \bullet C + C \bullet A \tag{4}$$

The evolution of adders starts with the RCA. They are easy to utilize and may be modified to any demanded dimension.

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However, they are relatively inefficient since carriers could be distributed across the entirety span of the adder [7]. RCA requires the output of the previous stage to operate the next stage, proving to be rather slow. To remove this redundancy, parallel adders have come to light. The CLA unlike RCA utilizes the parallel computation technique to generate a carry-out bit for each bit neglecting the carry-in of the previous bits. These were the subsequent widely recognized adders. They are among the group of high-speed adders discovered to date due to reduced critical path delay [8], [9]. Another extremely efficient adder type is the CSA [10]. These adders compute parallel addition with and without carry and a suitable outcome is based on the input carry. According to studies, optimized CSA can require up to 61.788% less transistor count than conventional CSA [11]. In contrast to ordinary adders, CSaA can store and partially carry findings to execute extensive multiplicative analysis with high speed along with improved delay time and dimension. These also use low power in terms of application [12], [13]. Lastly, CSkA is a prominent implementation of a long adder with low complexity and reduced delay. A high-speed adder with considerably reduced latency as the carry bit skips over blocks to transmit logic 1 to the result [14]. CSkA is highly efficient in terms of space, power consumption, and delay. Regarding standard designs, modified CSkA adder designs implementing concatenation and variable latency techniques have been shown to drastically save power and area [15].

As circuit complexity increases, new adder topology will be required to satisfy performance and power demands [16]. The contribution of this article is to exhibit a detailed description of different adders and a comprehensive analysis in terms of transistor count, average power, area, frequency, and propagation delay. This study not only highlights the strengths and limitations of each architecture but also provides a comprehensive understanding of the trade-offs between efficiency, speed, and area utilization.

II. Design and Methodology

The adders discussed in this study are designed using 90nm technology in Cadence Virtuoso. Beginning with the schematic design of the basic gate blocks including OR, AND, NOT, XOR, etc. The creation of each basic block requires transient analysis which shows the input and output behavior of the block as a function of time. As these are the basic building blocks of the adder circuits, it is crucial to make sure each block generates the correct logic according to its operation. Subsequently, the blocks are interconnected according to the adder circuit design and further transient analysis is conducted to ensure the correct adding function has been implemented. Fig. 1 shows the schematic diagram of 1-bit RCA. The schematic diagram of 4-bit RCA, CLA, CSA, CSaA, and CSkA are shown in Fig. 2, Fig. 3, Fig. 4, Fig. 5, and Fig. 6 consecutively. Functionality check of each adder is shown in Fig. 7, Fig. 8, Fig. 9, Fig. 10, and Fig. 11 as per the mentioned sequence. Additionally, Cadence Virtuoso Layout Suite is used for placement and routing. To ensure that the layouts are error-free and have followed the design rules of CADENCE, Layout vs. Schematic (LVS) check and Design Rule Check (DRC) are performed. The layout designs of all the adders are shown in Fig. 12, Fig. 13, Fig. 14, Fig. 15, and Fig. 16 sequentially.

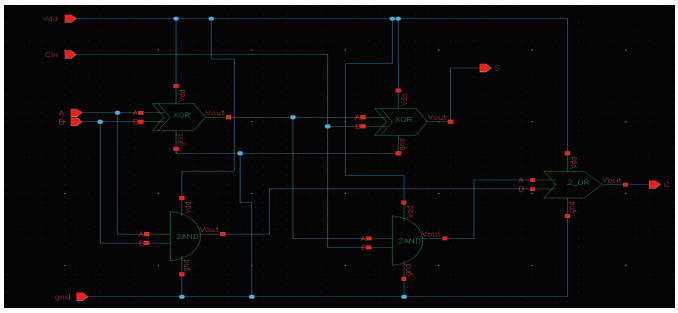


Fig. 1. 1-bit RCA schematic

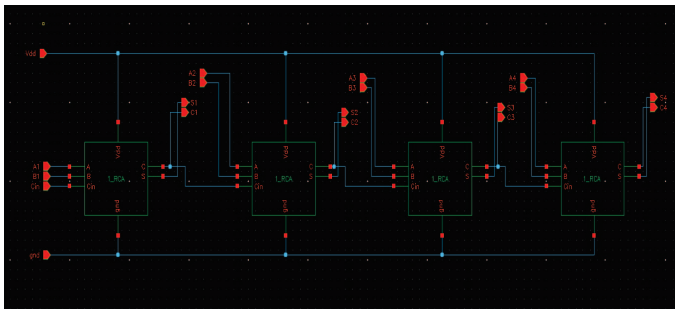


Fig. 2. 4-bit RCA schematic

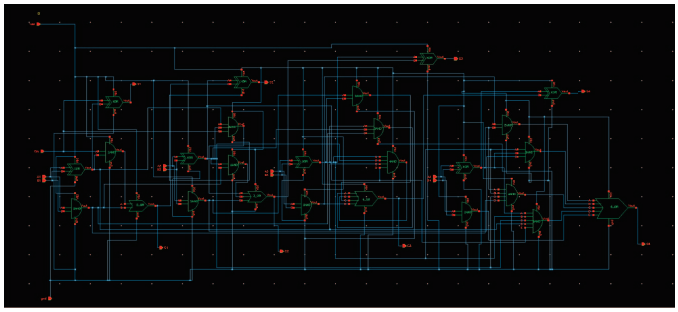


Fig. 3. 4-bit CLA Schematic

III. Result and Discussion

A set of performance metrics has been employed to compare and assess the simulated 4-bit adders. These include propagation delay, area, transistor count, frequency, and average power. The proposed 4-bit architectures' performance simulation results are summarized in Table I and Table II.

The comparative analysis reveals significant distinctions among the five types of adder architectures. To begin with, the CSkA demonstrates the lowest transistor count, contrasting with the CSA, which requires the highest count. This variation in transistor count arises from the fact that CSkA utilizes a multiplexer and parallel structure, allowing it for a lower

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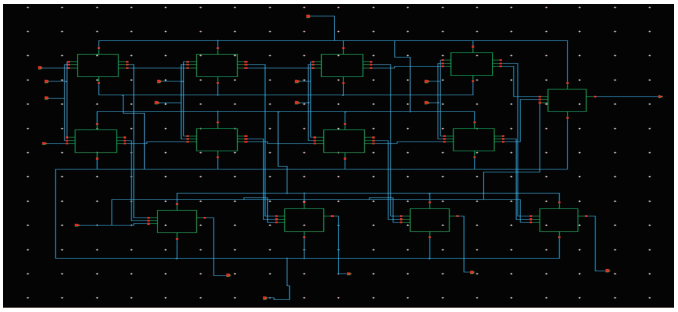


Fig. 4. 4-bit CSA Schematic

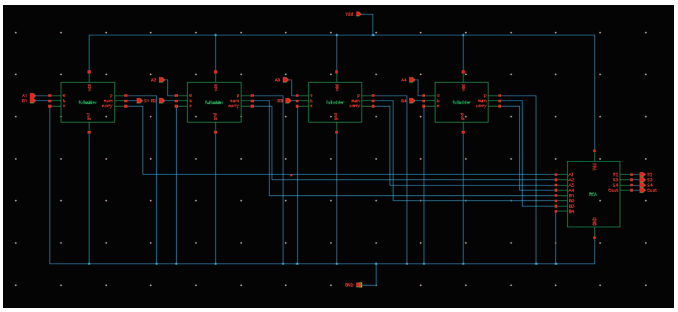


Fig. 5. 4-bit CSaA Schematic

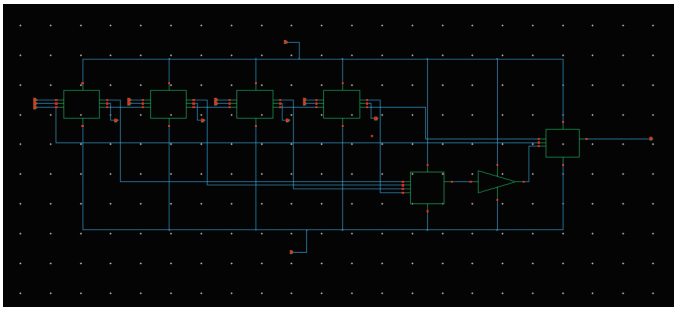


Fig. 6. 4-bit CSkA Schematic

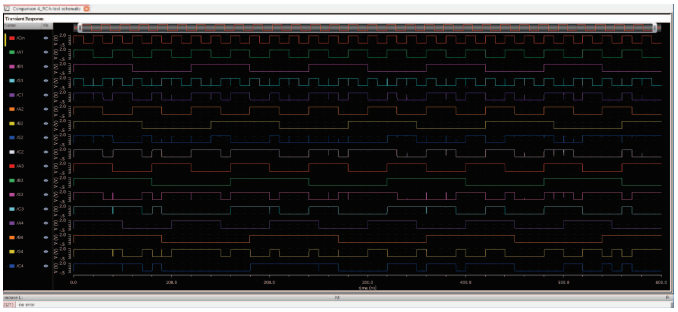


Fig. 7. 4-bit RCA Transient Analysis



Fig. 8. 4-bit CLA Transient Analysis

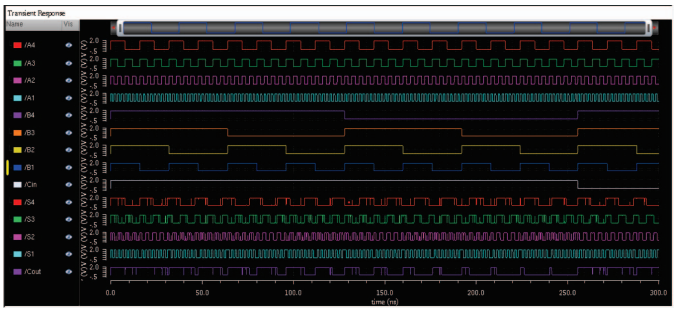


Fig. 9. 4-bit CSA Transient Analysis

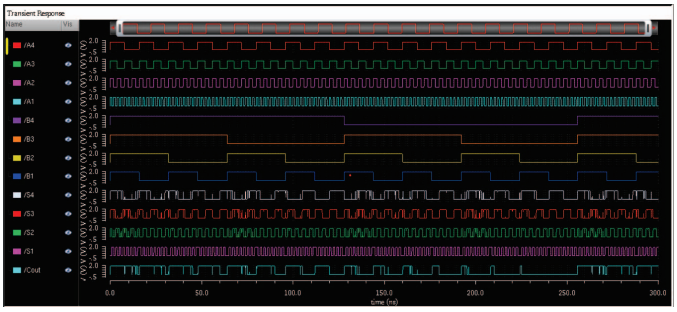


Fig. 10. 4-bit CSaA Transient Analysis

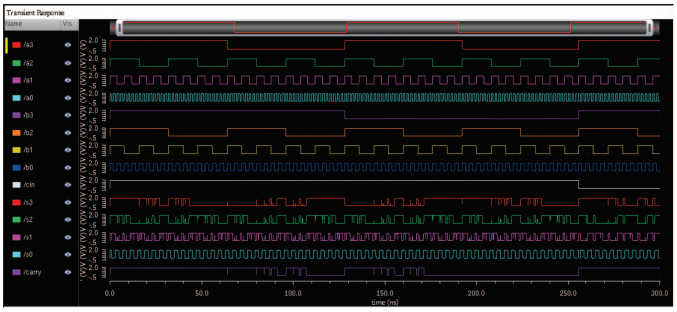


Fig. 11. 4-bit CSkA Transient Analysis

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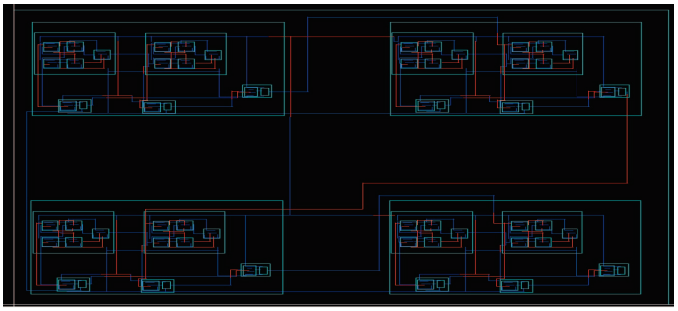


Fig. 12. 4-bit RCA layout

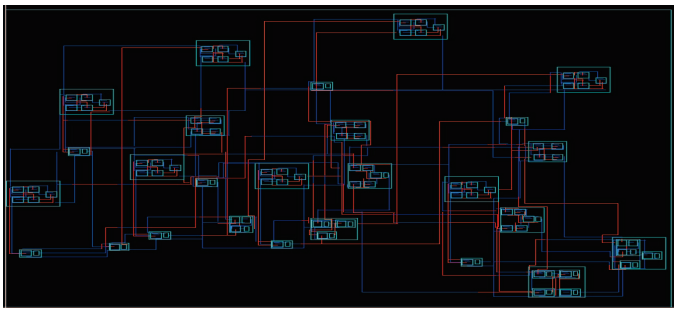


Fig. 13. 4-bit CLA Layout

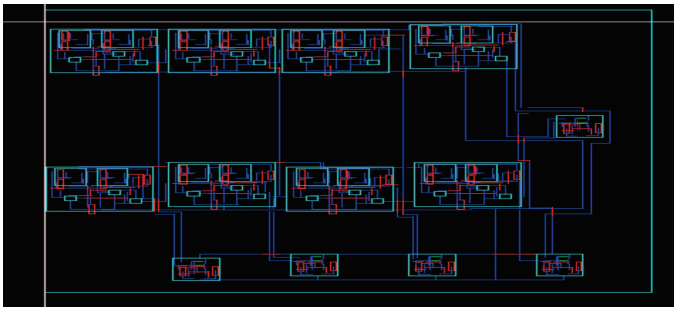


Fig. 14. 4-bit CSA Layout

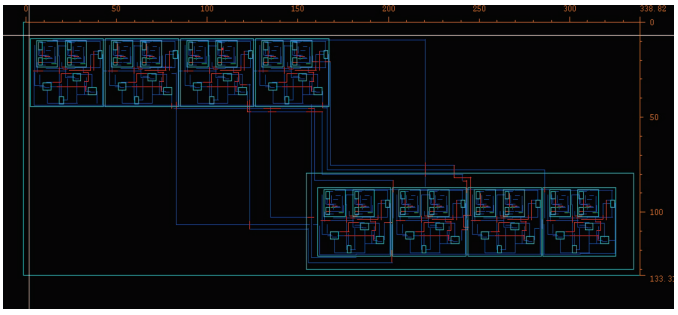


Fig. 15. 4-bit CSaA Layout

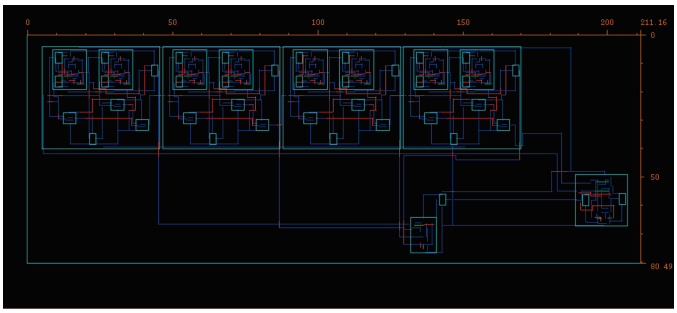


Fig. 16. 4-bit CSkA Layout

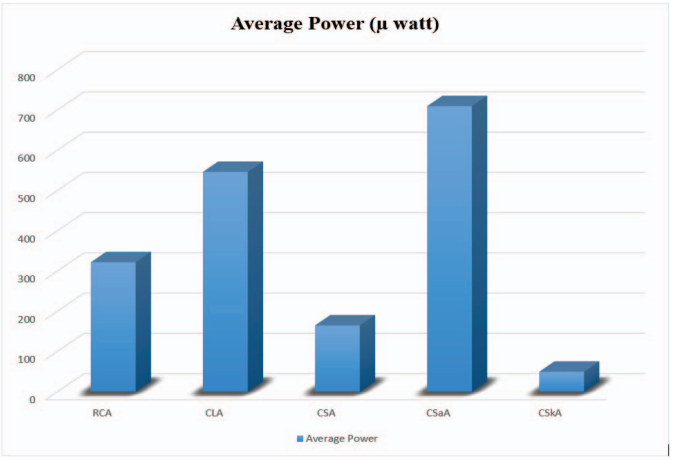


Fig. 17. Average Power Comparison

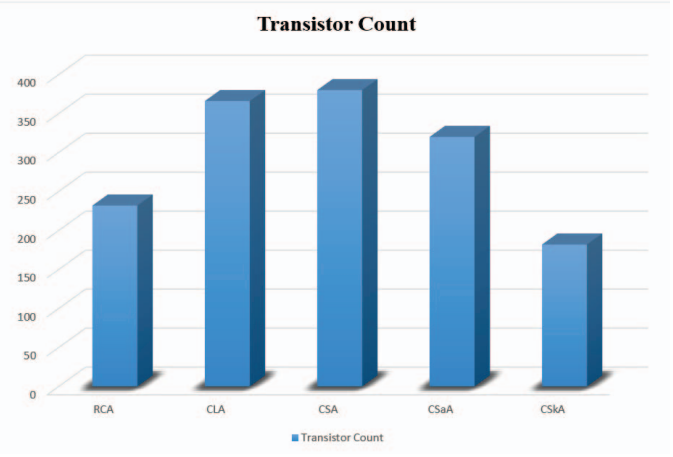


Fig. 18. Transistor Count Comparison

TABLE I
TRANSISTOR COUNT, AREA AND AVERAGE POWER COMPARISON OF 4-BIT ARCHITECTURES

4 Bit Adder	Transistor Count	Area (µm2)	Average Power (µ watt)
RCA	232	28420.75	322.09
CLA	366	72835.19	546.6
CSA	380	52900	165.1
CSaA	320	44954.48	709.9
CSkA	182	16880.26	50.04

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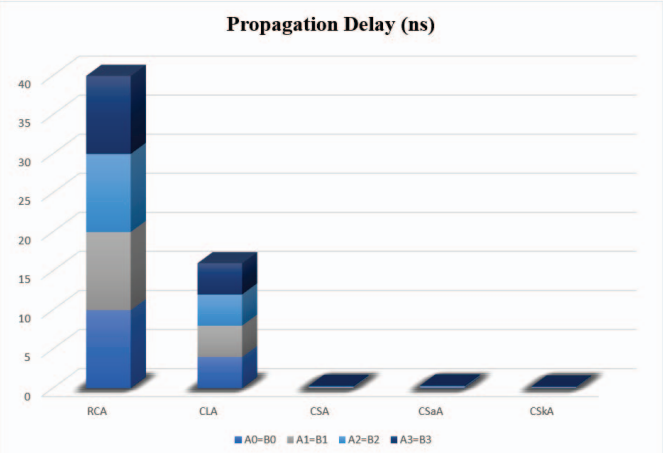


Fig. 19. Propagation Delay Comparison

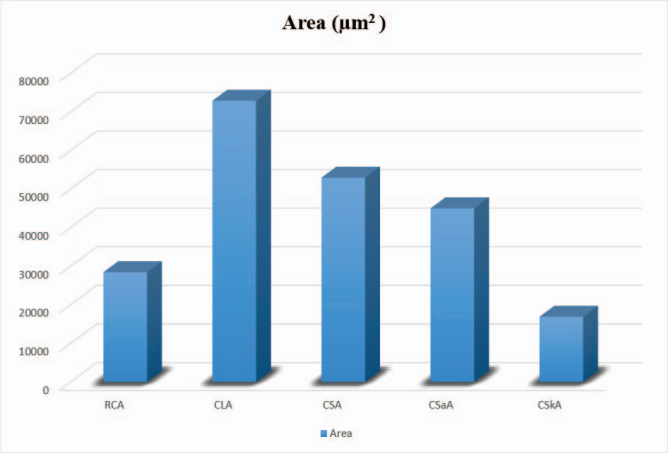


Fig. 20. Area Comparison

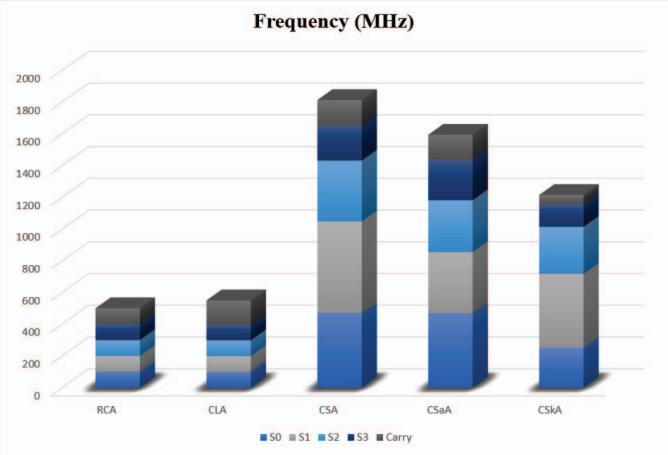


Fig. 21. Frequency Comparison

TABLE II
PROPAGATION DELAY AND FREQUENCY COMAPRISON OF 4-BIT ARCHITECHTURES

4 Bit Adder	Propagation Delay (s)	Frequency (MHz)
RCA	A_0=B_0=9.991n A_1=B_1=9.991n A_2=B_2=9.991n A_3=B_3=9.991n	F_{s0}=99.95 F_{s1}=99.65 F_{s2}=100.2 F_{s3}=100.146 $F_{carry-out}$ =100.146
CLA	A_0=B_0=3.998n A_1=B_1=3.998n A_2=B_2=3.998n A_3=B_3=3.998n	F_{s0}=99.74 F_{s1}=98.56 F_{s2}=24.97 F_{s3}=49.7 $F_{carry-out}$ =155.4
CSA	A_0=B_0=22.0062p A_1=B_1=95.6103p A_2=B_2=95.6466p A_3=B_3=94.635p	F_{s0}=471.5 F_{s1}=575.7 F_{s2}=383.6 F_{s3}=222.5 $F_{carry-out}$ =158.8
CSaA	A_0=B_0=85.9477p A_1=B_1=85.0464p A_2=B_2=85.0283p A_3=B_3=85.9533p	F_{s0}=469.8 F_{s1}=383.8 F_{s2}=327.6 F_{s3}=256.7 $F_{carry-out}$ =155.4
CSkA	A_0=B_0=25.565p A_1=B_1=64.6643p A_2=B_2=64.5809p A_3=B_3=64.572p	F_{s0}=252.5 F_{s1}=464.6 F_{s2}=296.3 F_{s3}=141.9 $F_{carry-out}$ =58.59

headroom requirement. Whereas CSA uses a higher number of transistors in its implementation.

Secondly, in terms of worst-case propagation delay, CSkA exhibits the shortest delay, a few picoseconds (ps). Skipping consecutive adder stages helps it to mitigate carry propagation delay without compromising the final carry-out. However, the opposite is seen in RCA with delays in the nanosecond (ns) range. Lacking parallel operation unlike other adders, the serial carry propagation extends the time-frame of the entire adder structure, with significant cumulative delay and dependency in previous stages.

CLA requires the largest layout area due to additional logic gates necessary for implementing the generate and propagate terms in the circuit, along with the look-ahead carry signals. Conversely, CSkA achieves the lowest average power consumption capability due to the skipping of intermediate carry blocks. Whereas, CSA requires the highest average power consumption due to the additional conversion steps required for the final sum.

Moreover, the carry save format of these adders requires additional power consumption. The last three adders demonstrate higher frequency, implying the parallel operation of these adders. Considering all the parameters, the CSkA emerges as the most efficient, characterized by the least average power consumption, compact area, and low transistor count along the lowest worst-case delay.

This study uses a 90nm Cadence tool to analyze five major adder circuits, evaluating several parameters across them. It includes schematic and layout design, DRC and LVS checking, making the findings more comprehensive and perceptive than previous studies. Comparisons with existing work are provided

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in Table III.

TABLE III
PERFORMANCE COMPARISON OF DIGITAL ADDER ARCHITECTURES IN CMOS STANDARD-CELL TECHNOLOGY

	Adders	Power Dissipation (μ watt)	Propagation Delay (sec)
Proposed Method	RCA	322.09	$9.9n$
	CLA	546.6	$3.398n$
	CSA	165.1	$22.0062p$
	CSaA	709.9	$85.9477p$
	CSkA	50.04	$25.564p$
Paper-1 [17]	RCA	1485	$19.66n$
	CLA	-	-
	CSA	-	-
	CSaA	1743	$2.228n$
	CSkA	6246	$125.5n$
Paper-2 [18]	RCA	-	$8.764n$
	CLA-HIER/CLA	-	$5498p$
	CSA	-	-
	CSaA	-	-
	CBYPASS-UNIF/CSkA	-	$5818p$
Paper-3 [19]	RCA	0.502	18.771
	CLA	-	-
	CSA	0.952	18.616
	CSaA	0.603	18.564
	CSkA	0.602	18.598

IV. CONCLUSION

The downsizing of VLSI technology has heightened the demand for optimum area utilization, minimal propagation delay, increased operating frequency, and enhanced power efficiency. As faster, more power-efficient digital devices emerge, addressing compactness and low power consumption requires standard CMOS technology evaluation based on performance factors. In this study, comparative analysis among the five mentioned adders is represented in bar diagrams in Fig. 17, Fig. 18, Fig. 19, Fig. 20, and Fig. 21, considering average power consumption, transistor count, propagation delay, area, and frequency respectively. From this study, it can be concluded that when all factors such as the lowest percentage of power consumption, area, transistor count, and worst-case delay are taken into account, CSkA is the most effective type of 4-bit adder. It is the most economical, energy-efficient, and time-saving adder available.

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